

EDS_ASSIGNMENT_02

NAME:

PRN:

BATCH:

DATASET: OpinionRank Review

PROBLEM 1: Display all records in dataset

```
import pandas as pd
import numpy as np

data = {
    "Review": [
        "Hotel was very good and clean",
        "Bad service and poor staff",
        "Excellent stay, loved it",
        "Average experience, nothing special",
        "Very bad, not recommended"
    ],
    "Rating": [5, 2, 5, 3, 1],
    "Hotel": ["A", "B", "C", "D", "E"]
}

df = pd.DataFrame(data)
df
```

	Review	Rating	Hotel
0	Hotel was very good and clean	5	A
1	Bad service and poor staff	2	B
2	Excellent stay, loved it	5	C
3	Average experience, nothing special	3	D
4	Very bad, not recommended	1	E

PROBLEM 2: Display first 3 records

```
df.head(3)
```

	Review	Rating	Hotel
0	Hotel was very good and clean	5	A
1	Bad service and poor staff	2	B
2	Excellent stay, loved it	5	C

Next steps: [Generate code with df](#) [New interactive sheet](#)

PROBLEM 3: Display last 2 records

```
df.tail(2)
```

	Review	Rating	Hotel
3	Average experience, nothing special	3	D
4	Very bad, not recommended	1	E

PROBLEM 4: Display all column names

```
df.columns
```

```
Index(['Review', 'Rating', 'Hotel'], dtype='object')
```

PROBLEM 5: Display data types of columns

```
df.dtypes
```

Review	object
Rating	int64
Hotel	object

dtype: object

PROBLEM 6: Display shape of dataset

```
df.shape
```

...

(5, 3)

PROBLEM 7: Display Review and Rating

```
df[["Review", "Rating"]]
```

...

	Review	Rating
0	Hotel was very good and clean	5
1	Bad service and poor staff	2
2	Excellent stay, loved it	5
3	Average experience, nothing special	3
4	Very bad, not recommended	1

PROBLEM 8: Display reviews having Rating greater than 3

```
df[df["Rating"] > 3]
```

...

	Review	Rating	Hotel
0	Hotel was very good and clean	5	A
2	Excellent stay, loved it	5	C

PROBLEM 9: Display total ratings sum

```
df["Rating"].sum()  
... np.int64(16)
```

PROBLEM 10: Calculate average Rating

```
df["Rating"].mean()  
... np.float64(3.2)
```

PROBLEM 11: Display maximum Rating

```
df["Rating"].max()  
... 5
```

PROBLEM 12: Add new column Review_Length

```
df["Review_Length"] = df["Review"].apply(len)  
df
```

```
...      Review  Rating  Hotel  Review_Length
```

0	Hotel was very good and clean	5	A	29
1	Bad service and poor staff	2	B	26
2	Excellent stay, loved it	5	C	24
3	Average experience, nothing special	3	D	35
4	Very bad, not recommended	1	E	25

Next steps:

[Generate code with df](#)

[New interactive sheet](#)

PROBLEM 13: Sort dataset decreasing Rating wise

```
df.sort_values(by="Rating", ascending=False)
```

...

	Review	Rating	Hotel	Review_Length
0	Hotel was very good and clean	5	A	29
2	Excellent stay, loved it	5	C	24
3	Average experience, nothing special	3	D	35
1	Bad service and poor staff	2	B	26
4	Very bad, not recommended	1	E	25



PROBLEM 14: Display review with highest Rating

```
df.loc[df["Rating"].idxmax()]
```

...

	0
Review	Hotel was very good and clean
Rating	5
Hotel	A
Review_Length	29

dtype: object

PROBLEM 15: Display review with lowest Rating

```
df.loc[df["Rating"].idxmin()]
```

...

Review	Very bad, not recommended
Rating	1
Hotel	E
Review_Length	25

dtype: object

PROBLEM 16: Count number of reviews

```
df["Review"].count()
```

... np.int64(5)

PROBLEM 17: Display reviews containing word 'good'

```
df[df["Review"].str.contains("good", case=False)]
```

...

	Review	Rating	Hotel	Review_Length
0	Hotel was very good and clean	5	A	29

PROBLEM 18: Descriptive statistics of numeric columns

df.describe()

	Rating	Review_Length
count	5.000000	5.000000
mean	3.200000	27.800000
std	1.788854	4.438468
min	1.000000	24.000000
25%	2.000000	25.000000
50%	3.000000	26.000000
75%	5.000000	29.000000
max	5.000000	35.000000

PROBLEM 19: Display reviews having Rating between 2 and 5

df[(df["Rating"] >= 2) & (df["Rating"] <= 5)]

	Review	Rating	Hotel	Review_Length
0	Hotel was very good and clean	5	A	29
1	Bad service and poor staff	2	B	26
2	Excellent stay, loved it	5	C	24
3	Average experience, nothing special	3	D	35

PROBLEM 20: Rename column 'Rating' to 'Review_Rating'



```
df.rename(columns={"Rating": "Review_Rating"}, inplace=True)
df
```

...

	Review	Review_Rating	Hotel	Review_Length
0	Hotel was very good and clean	5	A	29
1	Bad service and poor staff	2	B	26
2	Excellent stay, loved it	5	C	24
3	Average experience, nothing special	3	D	35
4	Very bad, not recommended	1	E	25

